DEEPAK TP

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Career Objective

Seeking a challenging role in a progressive organization where I can apply my skills in Artificial Intelligence and Machine Learning, enhance my technical expertise, and contribute to organizational growth.

Education

Manipal School of Information Science *Aug 2024 – Present*
Master of Engineering in Artificial Intelligence and Machine Learning (CGPA – 8.5)
G M Institute of Technology *Aug 2020 – Jun 2024*
Bachelor of Engineering in Artificial Intelligence and Machine Learning (CGPA – 7.6)

Work Experience

Research Intern | IIIT, Sri City *May2026 – Present*

- Researching and prototyping an Agentic AI-powered XR assistant that combines Vision-Language Models, spatial memory, RAG/web retrieval, and immersive AR/VR interfaces to deliver context-aware reasoning, real-time environmental understanding, and interactive guidance for navigation and task assistance.

AI-Intern | National Technical and Research Organization (NTRO) *Aug 2025 – Feb 2026*

- Engineered an **offline Retrieval-Augmented Generation (RAG)** system for technical document summarization using **Docling**-based extraction, **hybrid retrieval (BM25 + dense embeddings)**, and **FAISS**, improving contextual grounding and reducing LLM hallucinations.
- **Optimized LLM inference and serving** with **vLLM (paged attention)**, enabling **low-latency, high-throughput offline deployment** for secure, cost-efficient **GenAI/NLP workloads**.
- Operated within **air-gapped systems**, strengthening understanding of **secure infrastructure**, **network communication**, and **system-level constraints**.

Projects

- **Transformer-Based Time Series Classification**
  Developed a Transformer-based supervised learning model for patient disease classification using 24-hour minute-level temperature time-series data (1,440 observations). Conducted extensive EDA with statistical analysis, visualizations, and FFT-based frequency domain analysis to identify temporal patterns and improve model performance.
- **Intelligent Legal Document Retrieval and Case Summarization System**
  Built a legal document retrieval and case summarization system using RAG for context-aware responses across 1,000+ PDFs. Implemented semantic search with vector embeddings and LLMs, gaining experience in NLP, Transformers, and information retrieval.

Skills

Languages & Frameworks: Python, TensorFlow, PyTorch,PowerBI,AgenticAI,LangGraph,Pandas,MCP,LangChain
Relevant Coursework: Computer Vision, Machine Learning, Linear Algebra, Probability, Statistics, RL, DL,Git,AWS
Extra-curricular: Interested in studying **Business strategies** and **case studies,** Organized college fest "**MALLIKA**".
Languages known: English, Kannada, Hindi, Telugu.

Certifications

- **IBM AI Engineering** [ongoing].
- **Prompt Engineering with llama 2&3**. [Coursera].
- **Introduction To Industry 4.0 And Industry Internet of Things**. IIT Kharagpur Elite Swayam (NPTEL)